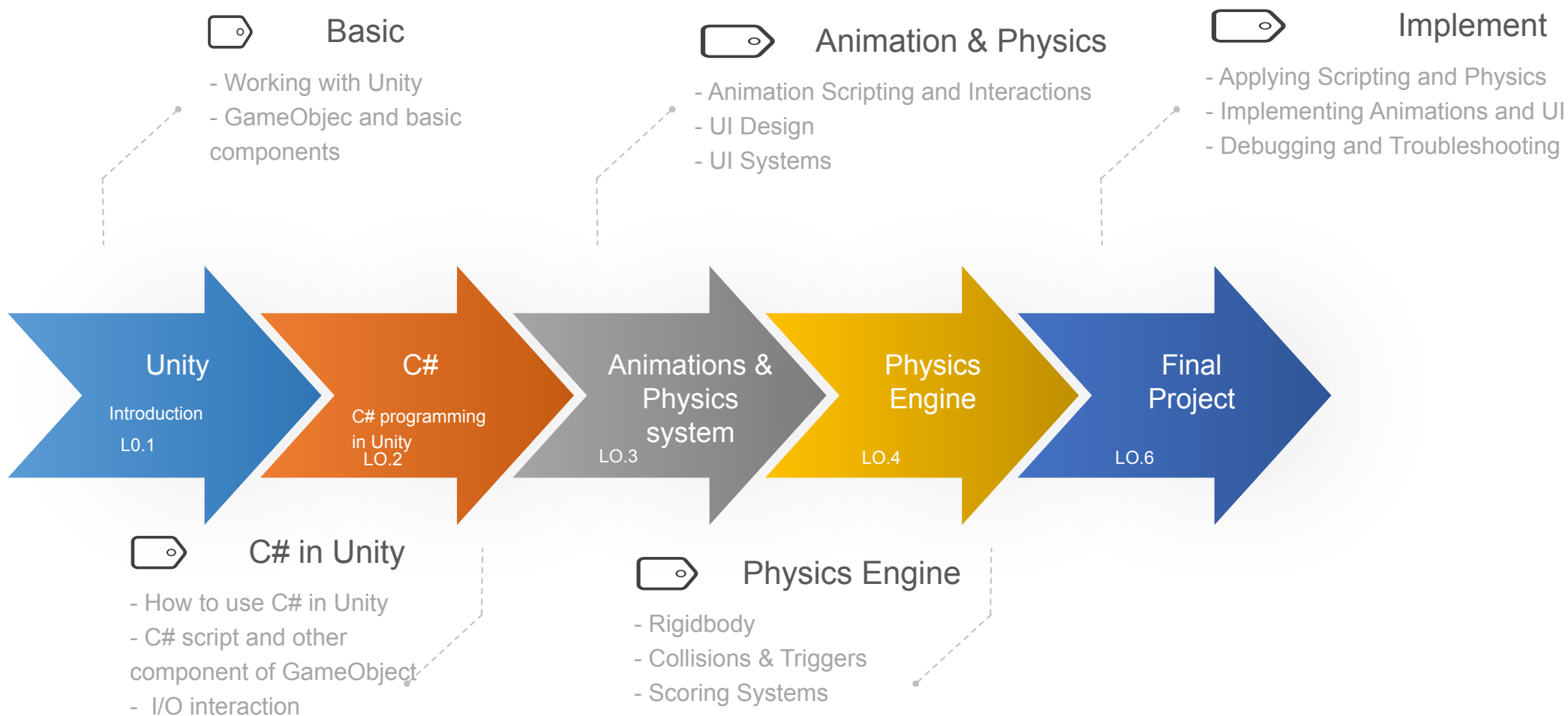




FPT UNIVERSITY

WELCOME TO UNITY

What you will learn in this Course



Using of Unity

- Game development
- Interactive Experiences
- Previsualization for Film
- Architectural Visualization
- Animation
- Simulations



FPT UNIVERSITY

Module 1

Introduction to Unity Interface and Navigation

Learning Objectives

- Familiarize with key components of Unity's interface.
- Learn efficient navigation techniques.
- Understand the role of various windows and tools.

Content

- Introduction to Unity
- Layout of Unity
- Unity Interface Components
- Unity's Scripting Integration
- Cross-Platform Capabilities

Introduction to Unity

- Unity is a professional-quality game engine used to create video games targeting a variety of platforms. Not only is it a professional development tool used daily by thousands of seasoned game developers, it's also one of the most accessible modern tools for novice game developers.

Introduction to Unity

- Unity is a cross-platform game engine initially released by Unity Technologies, in 2005. The focus of Unity lies in the development of both 2D and 3D games and interactive content. Unity now supports over 20 different target platforms for deploying, while its most popular platforms are the PC, Android and iOS systems.
- Unity features a complete toolkit for designing and building games, including interfaces for graphics, audio, and level-building tools, requiring minimal use of external programs to work on projects.

Layout of Unity

- Layout of Unity is designed to be both flexible and user-friendly, catering to the diverse needs of game developers. This layout is composed of several key panels or windows, which are introduced in the following slides, each serving specific functions that facilitate the development and testing of games.
- This layout is highly customizable, allowing users to drag and rearrange windows to suit their workflow preferences. Users can also save and switch between custom layouts depending on the task at hand, making Unity a versatile tool for game development across various stages of production.

Unity Interface Components



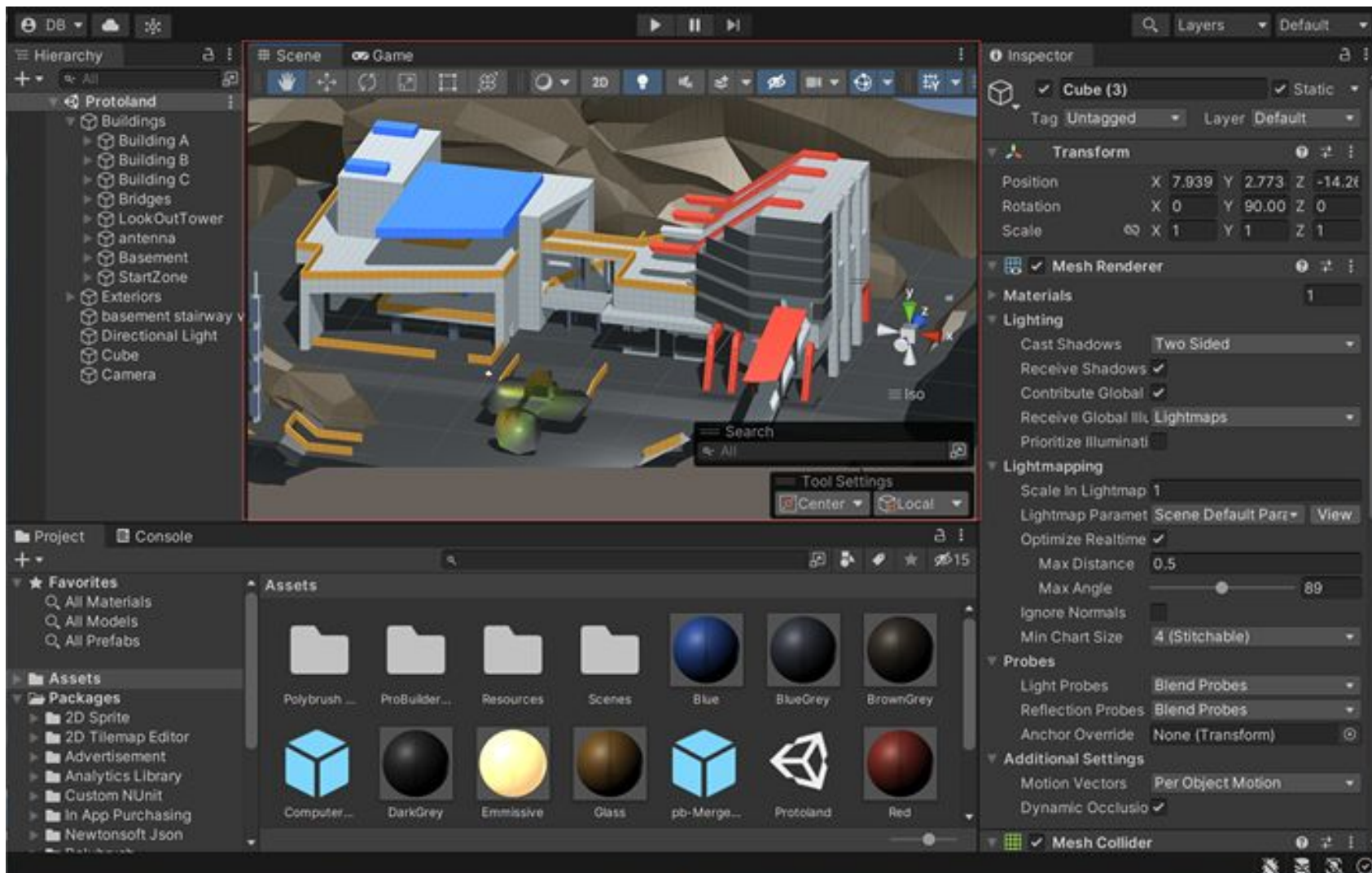
Unity Interface Components

- (A) [The Toolbar](#) provides access to your Unity Account and Unity Cloud Services. It also contains controls for Play mode; Undo history; Unity Search; a layer visibility menu; and the Editor layout menu.
- (B) [The Hierarchy window](#) is a hierarchical text representation of every GameObject in the Scene. Each item in the Scene has an entry in the hierarchy, so the two windows are inherently linked. The hierarchy reveals the structure of how GameObjects attach to each other.
- (C) [The Game view](#) simulates what your final rendered game will look like through your Scene Cameras. When you click the Play button, the simulation begins.
- (D) [The Scene view](#) allows you to visually navigate and edit your Scene. The Scene view can display a 3D or 2D perspective, depending on the type of Project you are working on.

Unity Interface Components

- (E) [Overlays](#) contain the basic tools for manipulating the Scene view and the GameObjects within it. You can also add custom Overlays to improve your workflow.
- (F) [The Inspector window](#) allows you to view and edit all the properties of the currently selected GameObject. Because different types of GameObjects have different sets of properties, the layout and contents of the Inspector window change each time you select a different GameObject.
- (G) [The Project window](#) displays your library of Assets that are available to use in your Project. When you import Assets into your Project, they appear here.
- (H) [The status bar](#) provides notifications about various Unity processes, and quick access to related tools and settings.

Scene View



Scene View

- **Purpose:** This is where you can visually manipulate and navigate your game world. It allows you to interact with and organize your game objects in 3D space or a 2D plane.
- **Features:** Tools for moving, rotating, and scaling objects; camera controls to view the scene from different angles; and options for toggling visibility and locking objects.

Scene View

- The Scene view is where you visualize and interact with the world you create in the Editor. In the Scene view, you can select, manipulate, and modify GameObjects that act as scenery, characters, cameras, lights, and more.

Topic	Description
Overlays	Manage the persistent, customizable panels and toolbars that contain Scene view authoring tools.
Position GameObjects	Modify the Transform values of GameObjects.
Scene view navigation	Move around the Scene view efficiently.
Scene view Camera	Configure the Scene Camera.
Pick and select GameObjects	Select GameObjects in the Scene view and manage their pickability.
Scene visibility	Hide and display GameObjects in the Scene view.
Scene view View Options toolbar	Select view options for the Scene View and enable or disable lighting and audio.
Gizmos menu	Hide and display gizmos in the Scene view.
Custom Editor tools	Create Editor tools to use in the Scene view.

Scene View Navigation

Use the View tool

When the View tool is selected (shortcut: **Q**), the following mouse controls are available:

Control	Description
Pan	      Click and drag to pan the Camera around.
Orbit	      Hold Alt (Windows) or Option (macOS), and left-click and drag to orbit the Camera around the current pivot point. This option isn't available in 2D mode, because the view is orthographic.
Zoom	      Hold Alt (Windows) or Option (macOS), and right-click and drag to zoom the Scene view. On macOS, you can also hold Control, and left-click and drag instead.

Hold down **Shift** to increase the rate of movement and zooming.

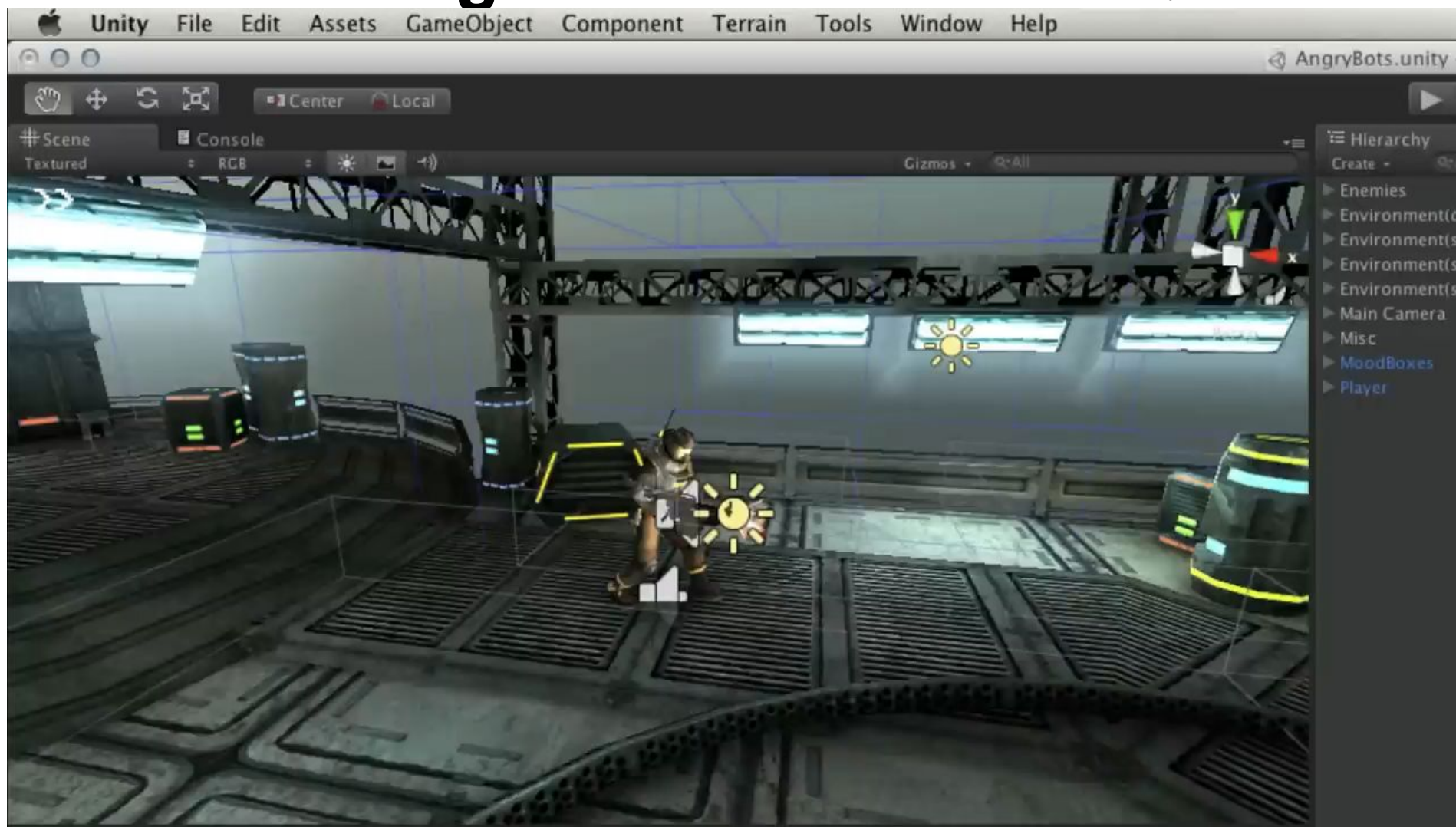
Movement shortcuts

For extra efficiency, these controls can also be used regardless of which transform tool is selected. The most convenient controls depend on which mouse or track-pad you are using.

Action	3-button mouse	2-button mouse or track-pad	Mac with only one mouse button or track-pad
Pan	Hold the middle-mouse button then drag	Hold Alt+Control +left-click, then drag	Hold Option+Command +left-click, then drag
Orbit (Not available in 2D mode)	Hold Alt +left-click, then drag	Hold Alt +left-click, then drag	Hold Option +left-click, then drag
Zoom	Use the scroll wheel, or hold Alt +right-click, then drag	Hold Alt +right-click, then drag	Use the two-finger swipe method to scroll in and out, or hold Option+Control +left-click, then drag
Change speed (only available in Flythrough mode)	Use the scroll wheel while moving.	Drag with two fingers while moving	Drag with two fingers while moving

Scene View Navigation

The Scene View - Unity Official Tutorials

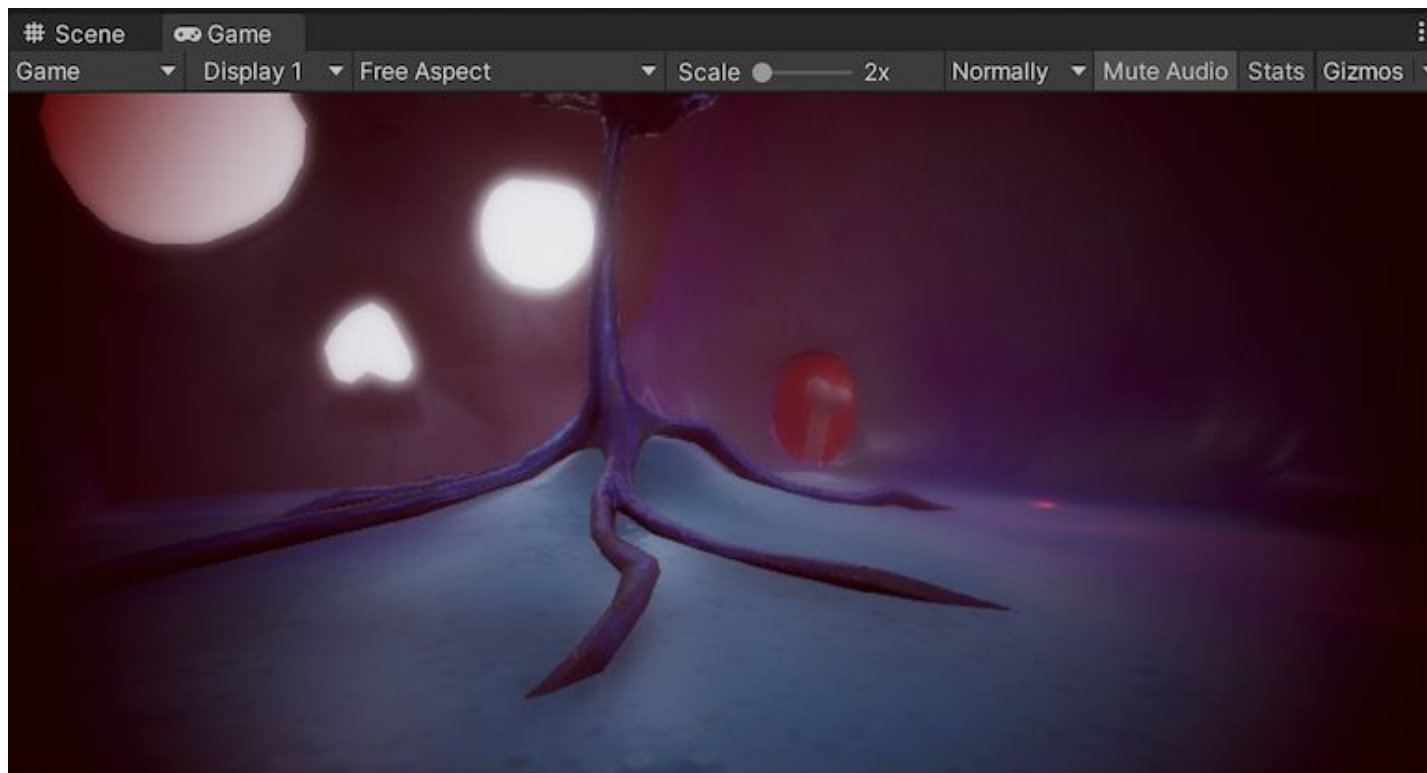


Game View

- **Purpose:** It provides a preview of what your game will look like to the end-user when it is running. This is the actual player's perspective.
- **Features:** Ability to play, pause, and step through your game right within the editor. It also displays the game from the camera's point of view.

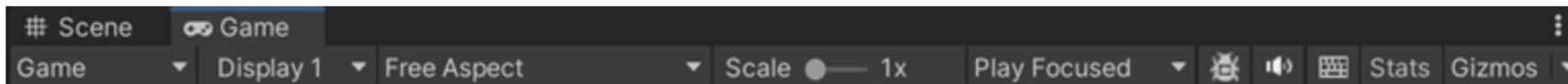
Game View

The Game view is rendered from the Camera(s) in your application. It shows how the final, built application looks. You need to use one or more Cameras to control what the player sees when they are using your application.



Game View

Game view Control Bar



Button	Function
Game/Simulator	Use this menu to enable the Game or Simulator view.
Display	If you have multiple Cameras in the Scene , select the appropriate option to switch between them. By default, this is set to Display 1 . You can assign Displays to Cameras in the Camera module, under Target Display .
Aspect ratio	Select from the available values to test how your game looks on monitors with different aspect ratios. By default, this is set to Free Aspect .
Low Resolution Aspect Ratios	Enable Low Resolution Aspect Ratios to emulate the pixel density of older displays, which reduces the resolution of the Game view when an aspect ratio is selected. It is always enabled when the Game view is on a non-Retina display.
Scale slider	Scroll right to zoom in and examine areas of the Game screen in more detail. This slider lets you zoom out to see the entire screen where the device resolution is higher than the Game view window size. You can also use the scroll wheel and middle mouse button to do this while the application is stopped or paused.

Game View

Play Mode behavior		This section describes the play mode behavior based on your selection below.
	Focused	Enable Focused to shift focus on the selected Game view while the Editor is in Play mode. Only one Game view can be in focus when you enter the Play mode. Using Maximized mode implies focus on the Maximized Game view. Enabling Focused on a Game view disables it on other Game views.
	VSync (Game view only)	Enable VSync (Game view only) to allow syncing, which is useful when recording a video, for example. Unity attempts to render the Game view at the monitor refresh rate, though this is not guaranteed. When this option is enabled, it is still useful to maximize the Game view in Play mode to hide other views and reduce the number of views that Unity renders.
Enter Play Mode:		Choose from the options below to determine the settings for the Editor when it enters the Play mode.
	Normally	Select this to view the Game view without forcing focus or maximizing any views to full screen.
	Maximized	When this option is selected, Unity runs the Play mode with the Game view maximized to 100% of the Editor window.
Mute audio		Click to enable: use this to mute any in-application audio when you enter the Play mode.
Stats		Click this to show or hide the Statistics overlay, which contains Rendering Statistics about your application's audio and graphics. The overlay lets you monitor the performance of your application while in Play mode.
Gizmos		Click this to show or hide the visibility of Gizmos . To only show certain types of Gizmos in the Play mode, from the Gizmos menu, only enable the Gizmo types you want to see.

Hierarchy Window

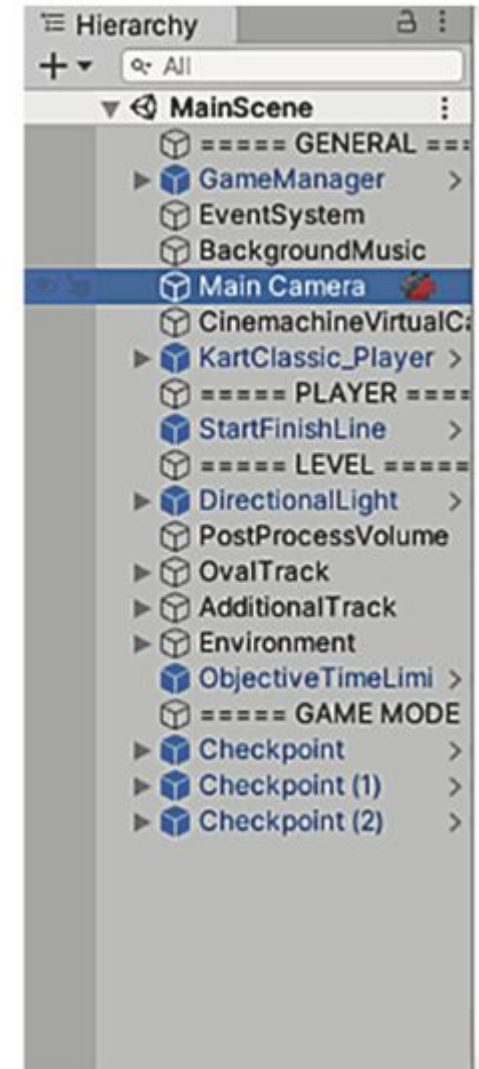
- **Purpose:** Shows all the game objects in the current scene, both active and inactive. It represents the parent-child relationship between objects, which is crucial for organizing complex scenes.
- **Features:** Easy reordering, searching, and nesting of objects. You can also right-click to create new objects or make them children of others.

Hierarchy Window

Hierarchy lists the name of every object in the scene and nests the names together according to their hierarchy linkages in the scene.

Basically, it's a way of selecting objects by name instead of hunting them down and clicking them within the Scene view.

The Hierarchy linkages group objects together visually, like folders, allowing you to move the entire group as one

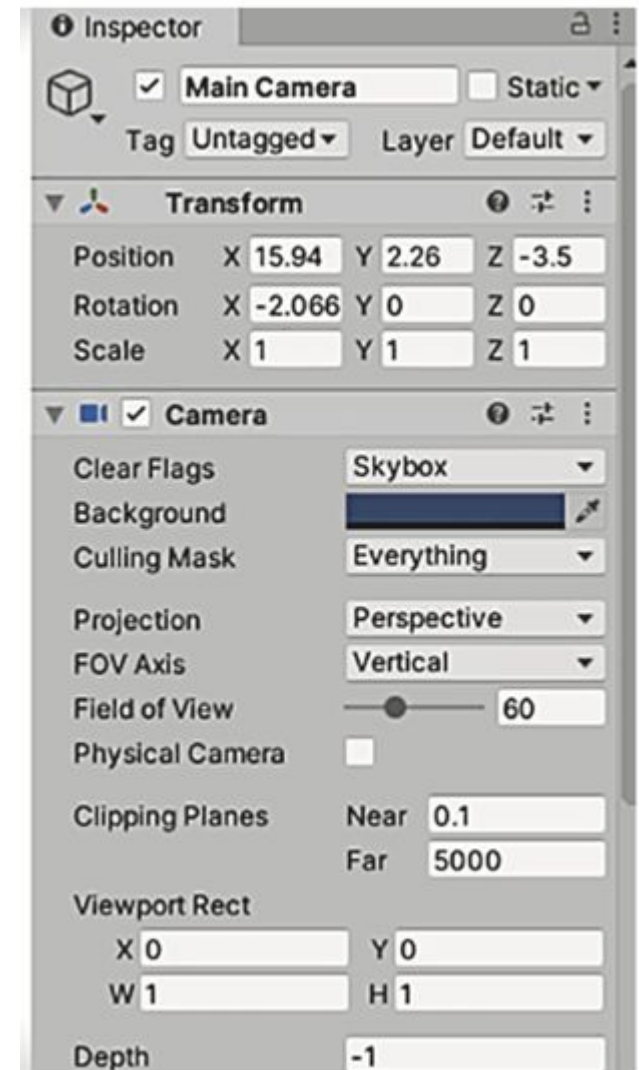


Inspector Window

- **Purpose:** Displays detailed information about the currently selected object. You can view and edit all properties of a game object here, including components, materials, scripts, and physics properties.
- **Features:** Add new components or scripts; edit properties like color, size, and physical behaviors; apply changes to prefab instances.

Inspector Window

The Inspector shows you information about the currently selected object. Select an object, and the Inspector is then filled with information about that object. The information shown is pretty much a list of components, and you can even attach or remove components from objects.



Toolbar

- **Purpose:** Provides quick access to tools for manipulating the scene and game objects, and controls for playing, pausing, and stepping through the game.
- **Features:** Transform tools (move, rotate, scale); play controls; layout selector to switch between custom and pre-set window layouts.

Toolbar

You can find the Toolbar at the top of the Unity Editor. It is not a window, and is the only part of the Unity interface that you can't rearrange.



Control:	Description:
	Access your Unity Account from the Account drop-down menu.
	Click the Cloud button to open the Unity Services window.
	Use the Play, Pause, and Step buttons in the Game view .
	Use Undo History to view, undo, or redo actions performed in the Editor
	Open Search .
	Control which objects appear in Scene view from the Layers drop-down menu.
	Change the arrangement of your views and then save the new layout or load an existing from the Layout drop-down menu.

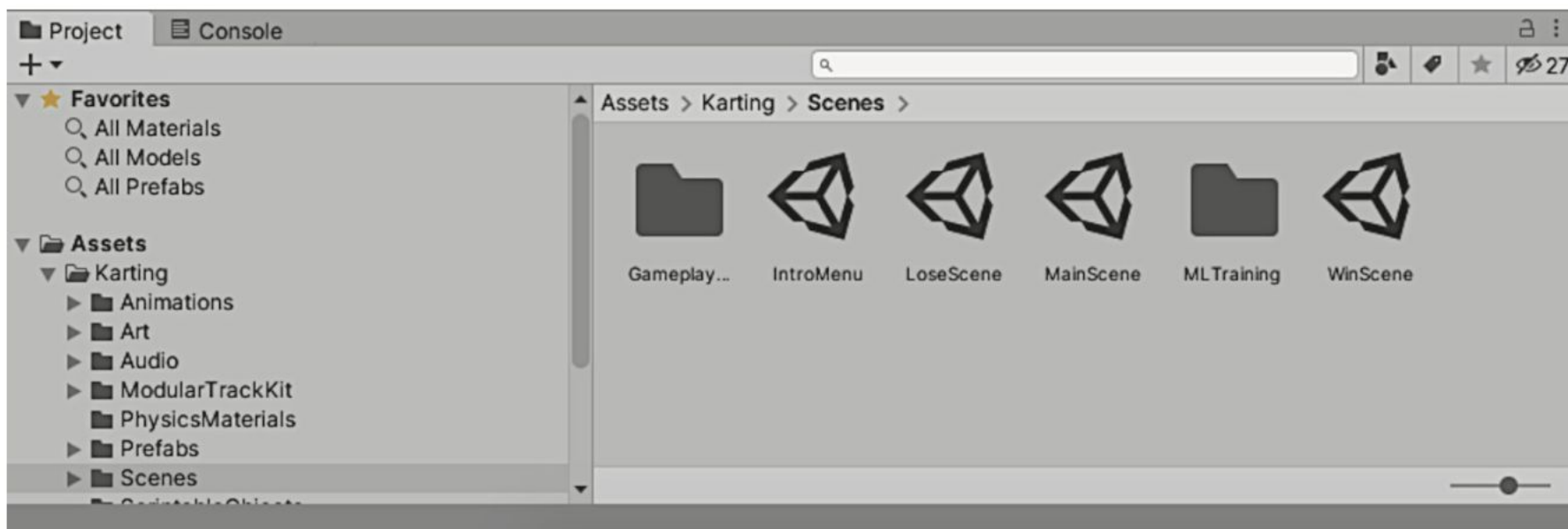
Project Window

- **Purpose:** This is essentially your file browser within Unity where you manage all assets associated with your project, such as scenes, scripts, materials, and textures.
- **Features:** Organize assets in folders; import new assets; view and edit asset properties; drag and drop assets into the scene.

Project window

Project shows all the assets (art, code, and so on) in the project. Specifically, on the left side of the view is a listing of the project's directories; when you select a directory, the right side of the view shows the individual files in that directory.

Project shows files that may not be contained within any specific scene.

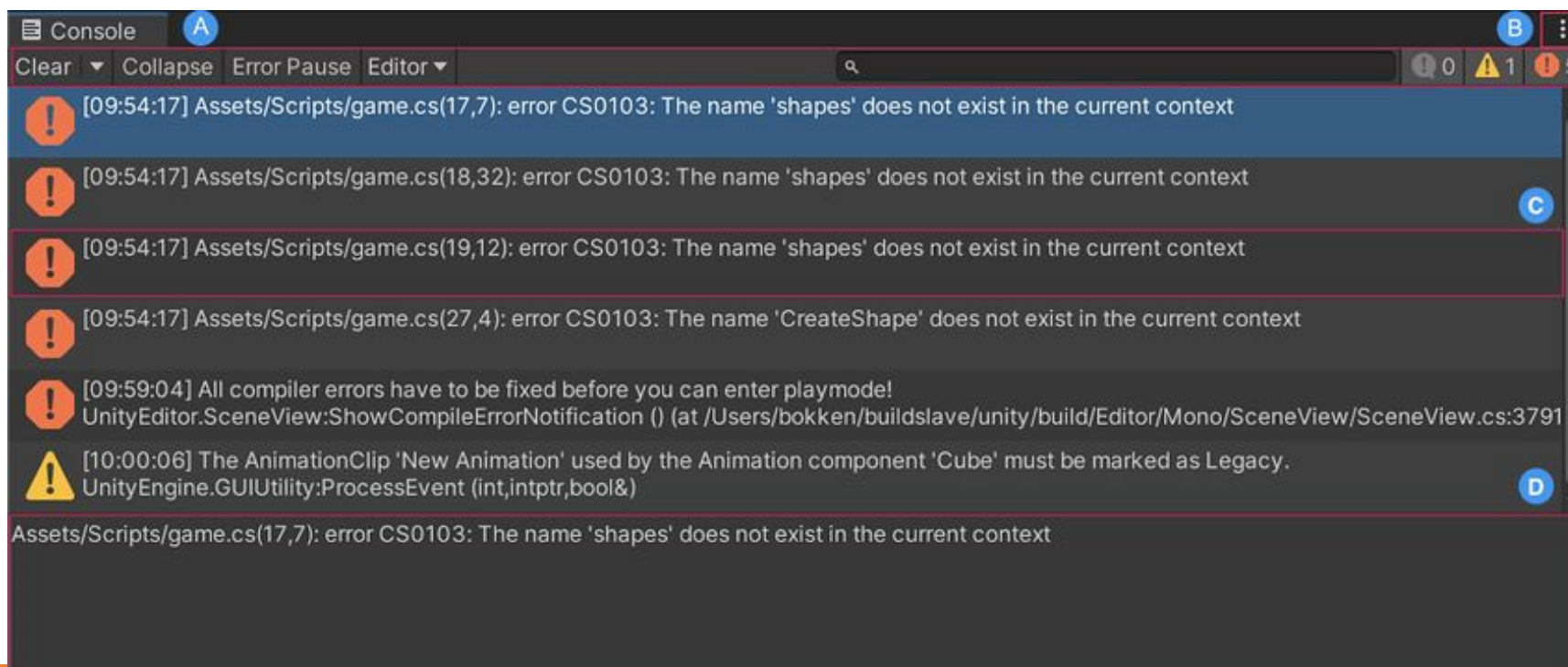


Console Window

- **Purpose:** Displays errors, warnings, and other messages generated by Unity. It is essential for debugging.
- **Features:** Filter messages; clear output; double-click to locate the source of errors in scripts.

Console window

The Console window is the place where messages from the code show up. Some of these messages will be debugging output that you placed deliberately, but Unity also emits error messages if it encounters problems in the script you wrote.

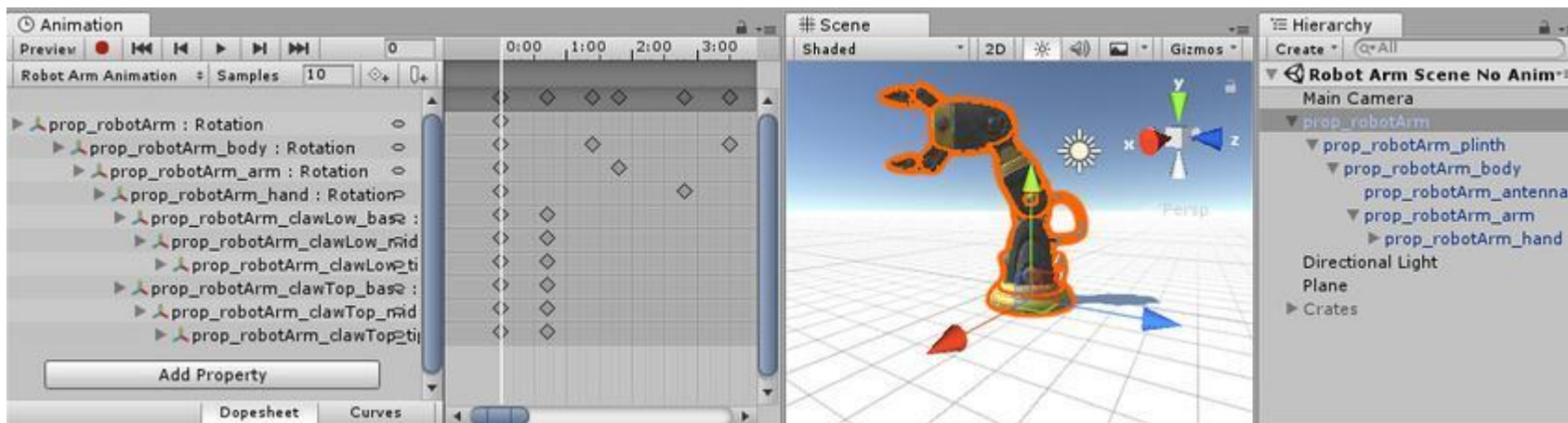


Animation and Animator Windows

- **Purpose:** These are used to create and manage animations for objects. Animation Window is for keyframe-based animation, while Animator Window manages states and transitions for animation flow.
- **Features:** Timeline for keyframes; curve editor for smooth transitions; state machine setup for complex animations.

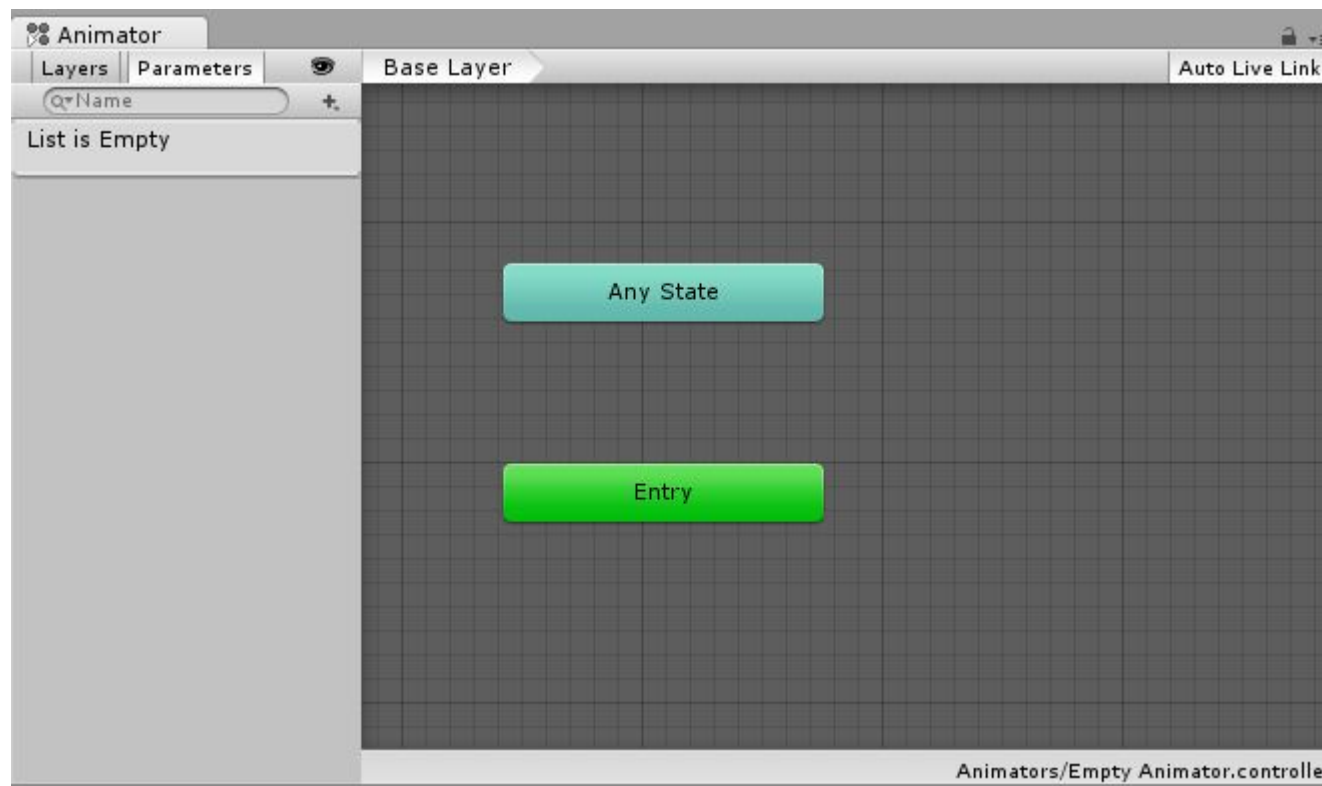
Animation and Animator Windows

- The **Animation** window is linked to the **Hierarchy** window, the **Project** window the **Scene** view, and the **Inspector** window. Like the **Inspector**, the **Animation** window shows the timeline and keyframes of the Animation for the currently selected GameObject or Animation Clip Asset. You can select a GameObject using the **Hierarchy** window or the **Scene** View or select an Animation Clip Asset using the **Project** Window.



Animation and Animator Windows

- The Animator Window allows you to create, view and modify Animator Controller assets.



Unity's Scripting Integration

Unity is deeply integrated with various scripting languages, with C# being the primary language used for scripting game logic within the Unity ecosystem.

- Scripts in Unity play a pivotal role in adding interactivity to game objects.
- These scripts define the behavior, interactions, and logic behind various game elements, making them dynamic and responsive to player input or events.

Scripts act as the backbone for game interactivity, enabling developers to define how game elements behave and respond in different situations or conditions.

Cross-Platform Capabilities

Unity's cross-platform capabilities empower developers to create games and applications that can seamlessly run on multiple platforms, including mobile devices (iOS, Android), consoles, desktops (Windows, macOS), web browsers, AR/VR devices, and more.

Unparalleled platform support

iOS

android 

 Windows





PS4



 XBOX ONE



androidtv

tvOS



 ARCore

 Microsoft HoloLens

 magic leap

 Meta



Unity Community and Support

Unity Community

Forum: <https://forum.unity.com/>

Discussions:

<https://discussions.unity.com/>

Community: <https://unity.com/community>

Unity Manual

<https://docs.unity3d.com/> (*)

<https://learn.unity.com/>

Unity Asset Store

<https://assetstore.unity.com/>

Summary

- **Understanding Unity's Interface:** Explored the fundamental components—Scene view, Game view, Hierarchy, Project panel, Inspector, and Toolbar—underlining their roles in the game development workflow.
- **Panel and Tool Roles:** Emphasized the significance of each panel and tool, detailing their contributions to game creation, manipulation, and asset management.
- **Scripting and Cross-Platform Capabilities:** Introduced Unity's scripting integration, highlighting the role of scripts in adding interactivity, and discussed Unity's cross-platform adaptability for diverse deployment.

References

- 1: Hocking, Joseph; Schell, Jesse, Unity in action: multiplatform game development in C#, 2022
- 3: Geig, Mike, Sams teach yourself Unity Game development in 24 hours, 2014
- 8: Unity Technologies, Unity Manual, 2023
- 9: Tutorials Point, Learning Unity, 2023